

DOCUMENTATION

Assumptions

Defining “**on-screen dialogue**” – it is primarily the text that shows up on the screen

It could be a video frame where it shows up

It could be a hardcoded subtitle in the given URL

If there is an audio aspect of the dialogue, that is noted and the timestamp of that particular dialogue.

Approach:

I used a polyglot architecture for building this system.

Whisper for extracting from the audio

RapidOCR (ONNX Runtime) used for extracting frame – given the constraints on my laptop, 8 GB RAM & AMD Ryzen 5 5500u Processor

This was a lightweight option

Can be replaced with better and stronger DL models

Traversal

1. Starts by gathering the metadata of the video, followed by checking the audio aspects, if it catches the text to be searched; it stops and checks if any frames match the required text from 00:00:00 to the current timestamp

This acts like a two-pointer approach

Why not just check near the occurred timestamp?

The basic idea is to find the first instance of the dialogue.

So, checking from the start ensure this is handled.

1. When the pointers reach at the same instance, it implies, it has not found the frame or the audio in that time range. This loop continues.
2. The process terminates as soon as the criteria is met.
 - a. Priority 1 – Video frame found – return required details (image, timestamp)
 - b. Priority 2 – Audio found, but no image – return the timestamp
 - c. Priority 3 - No video, no audio – return apt message
3. **How you handle cases where the result is ambiguous or uncertain**
The processing checker – we use Fuzzy Logic to check for how accurate the Frame could be (Levenstein Distance)
4. The accuracy threshold is set to 85%
 - a. It can be manually updated in the front end
 - b. The range is 60 – 100
 - c. A poor-quality video should be set to a lower threshold & vice versa for a higher quality video
5. VirtualThreads in Java enable a proper streamlined approach
6. Zombie processes are taken care of
7. Termination of failed threads and other instances are taken care under the ACID properties, proper roll-back and re-checking is implemented
8. Binary search is done on the frame.

Challenges Faced

1. Facing rate limits on the “ok.ru” website (can be overcome with the help of a VPN {I mostly swapped between mobile hotspot and college Wi-Fi})
2. The content of the code was vague and bringing it to match the problem statement was critical and I was able to get into the nitty-gritties of the code and architect the solution as per my thought process
3. Computationally weak Laptop, just took a bit longer time to run

Core Logic of the frame testing

1. It takes a single paused frame from the video, system uses **CLAHE**, cleans up the lighting or shadows, and scans it for written words.
2. It first checks if your target phrase is written perfectly on a single line
3. If it cannot find it on one line, it gathers every single word scattered across the screen into one giant clump and checks again.
4. This gathering technique ensures the system can still read the words even if they are broken up, animated, or scattered across the screen
(It cannot handle curved texts and complex emojis)

Polyglot Architecture

1. Java's Strength: Java is world-class at enterprise orchestration, memory management, and high-speed concurrency (handling multiple tasks at once without slowing down).
2. Python's Strength: Python has the most advanced, ready-to-use ecosystem for Artificial Intelligence, Machine Learning, and Computer Vision.
3. The Synergy: Managing heavy, multi-threaded video chunking purely in Python, the system would likely bottleneck. Java running ML models would be a nightmare. A polyglot system gives me Java's muscle and Python's brains.
4. In terms of Isolation
 - a. Preventing Cascading Failures: In a single-language application, if an intense ML process runs out of memory, it brings down the entire web server with it.
 - b. Safe Boundaries: Because the system separates the logic, they act independently. Example - Python extraction failed due to a codec error, the Java orchestrator stayed perfectly alive, caught the error, and kept the user interface running smoothly.
5. Independent Scalability - AI processing requires massive amounts of CPU and memory, whereas routing web traffic requires very little. Helps me scale both individually
6. Producer – Consumer Pattern
 - a. Video Tool (Producer) chops the video into chunks of frames
 - b. Search engine picks up the videos and feeds it to the model
7. Strategy Pattern – Java code is independent of the core logic
Tomorrow, I can swap the RapidOCR with a stronger DL algorithm.

Examples:

Link - <https://www.youtube.com/watch?v=9FnO3igOkOk>

Text - You cant handle the truth (SPOKEN, BUT NOT VISIBLE ON SCREEN)

```
[10:26:14 PM] Initiating connection to Java orchestrator...  
[10:26:14 PM] Job accepted. Assigned ID: 65be77f2-67a7-4e64-b732-0adc64393227  
[10:26:16 PM] [INITIALIZING] Extracting raw stream URL from https://www.youtube.com/watch?v=9FnO3igOkOk  
[10:26:21 PM] [INITIALIZING] Fetching stream metadata via ffprobe...  
[10:26:22 PM] [AUDIO_SEARCH_RUNNING] Launching Audio Pointer and Coarse Visual Scan...  
[10:27:25 PM] [AUDIO_EXTRACTING_FRAME] Fetching visual context for audio match...  
[10:27:30 PM] [SUCCESS_AUDIO] Text not found visually. Audio match located.
```

Extraction Result

Timestamp: 00:00:44.500

Frame Number: 1067 (Audio Match Context)

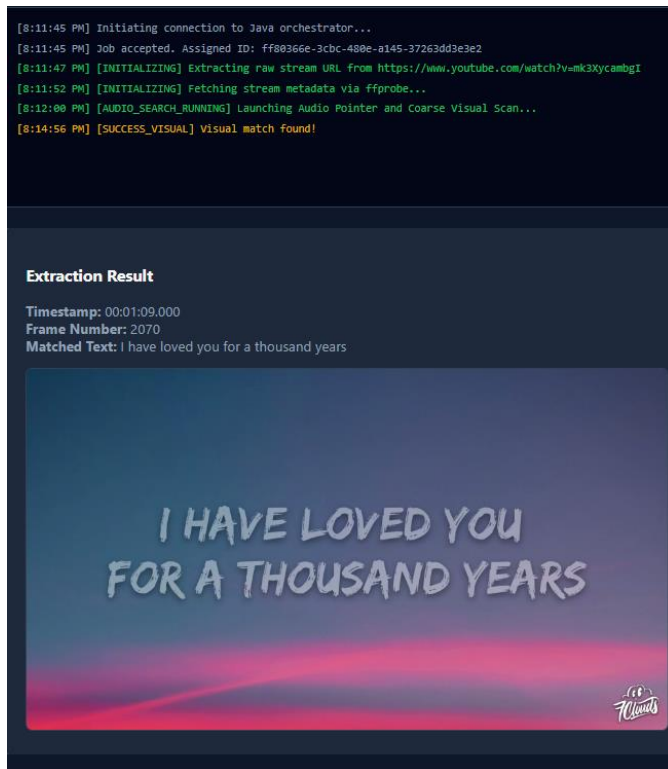
Matched Text: You can't handle the truth

Target text was spoken, but did not appear visually on screen.



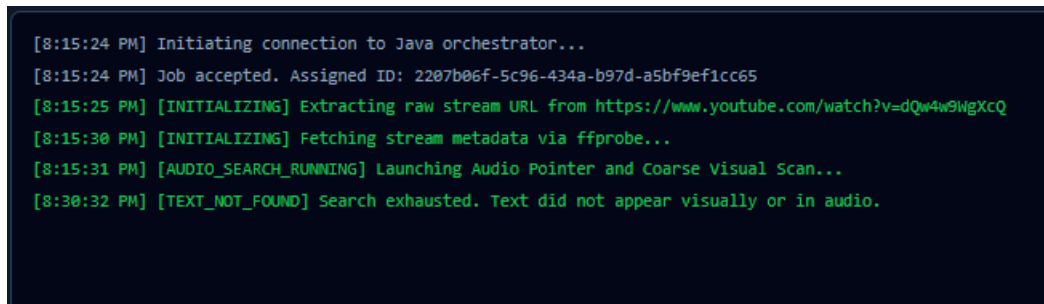
Link - <https://www.youtube.com/watch?v=mk3Xycambgl>

Text - I have loved you for a thousand years (TEXT IN THE FRAME CASE)



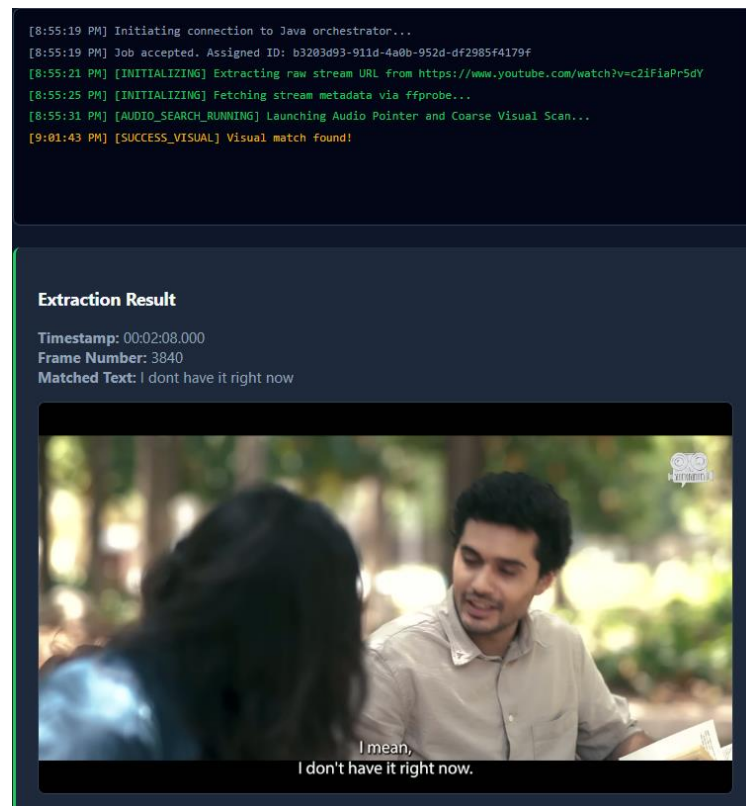
Link - <https://www.youtube.com/watch?v=dQw4w9WgXcQ>

Text - My Mind Rebels at Stagnation (NO TEXT OR AUDIO MATCH CASE)



Link - <https://www.youtube.com/watch?v=c2iFiaPr5dY>

Text - I dont have it right now (SUBTITLES CASE)



Link - <https://ok.ru/video/248244667877>

Text - My mind rebels at stagnation

(HIT TOO MANY REQUESTS TO THE SITE, GOT MYSELF TIMED OUT)
(COULD NOT GENERATE AN IMAGE, WILL UPDATE)